```
**************************************************** 
*                                                  *
*           STAAD.Pro V8i SELECTseries4            *
*           Version  20.07.09.21                   *
*           Proprietary Program of                 *
*           Bentley Systems, Inc.                  *
*           Date=    MAY  6, 2014                  *
*           Time=    21:38:19                      *
*                                                  *
*      USER ID: Parsons Brinckerhoff               *
****************************************************
```

```
     1. STAAD SPACE
INPUT FILE: 140506_ASF_Kallang.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 21-JAN-13
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 1 0 -0.415 0; 2 0 0 0; 3 0 -0.415 6.1; 4 0 0 6.1; 5 0 1.8 0; 6 0 1.8 6.1
     9. 7 0 4.3 0; 8 0 4.3 6.1; 9 0 6.1 0; 12 0 0 4.3; 15 0 6.1 4.3; 17 0 0 1.8
    10. 20 0 6.1 1.8; 24 3.33 -1.2 6.1; 26 4.95 -1.2 6.1; 30 5.68 -1.2 1.8
    11. 31 5.68 -1.2 4.3; 32 5.5 -1.2 0; 33 5.5 -1.2 6.1; 34 3.33 -1.2 0
    12. 36 4.95 -1.2 0; 37 0 6.1 6.1; 38 0 6.484 0; 39 0 6.484 6.1
    13. MEMBER INCIDENCES
    14. 1 1 2; 2 3 4; 3 2 17; 6 9 20; 7 4 6; 8 6 8; 10 2 5; 11 5 7; 13 12 4; 19 15 37
    15. 21 17 12; 27 20 15; 29 5 17; 30 6 12; 31 20 7; 32 15 8; 35 6 24; 37 8 37
    16. 38 8 26; 43 12 31; 44 17 30; 46 37 33; 47 9 32; 48 5 34; 49 7 9; 50 7 36
    17. 51 9 38; 52 37 39
    18. START USER TABLE
    19. TABLE 1
    20. UNIT METER KN
    21. WIDE FLANGE
    22. H914
    23. 0.0740918 0.9601 0.031 0.4275 0.0559 0.0113584 0.000730001 5.82068E-005 -
    24. 0.0297631 0.0477945
    25. H400
    26. 0.05244 0.458 0.03 0.417 0.05 0.00185878 0.00060507 3.7972E-005 0.01374 0.0417
    27. END
    28. DEFINE MATERIAL START
    29. ISOTROPIC STEEL
    30. E 2.05E+008
    31. POISSON 0.3
    32. DENSITY 76.8195
    33. ALPHA 1.2E-005
    34. DAMP 0.03
    35. END DEFINE MATERIAL
    36. CONSTANTS
    37. BETA 90 MEMB 3 6 13 19 21 27 29 TO 32
    38. MATERIAL STEEL ALL
```

```
39. MEMBER PROPERTY BRITISH
40. 35 38 48 50 TABLE ST UC356X406X393
41. 43 44 46 47 TABLE ST UC356X406X287
42. MEMBER PROPERTY AMERICAN
43. 3 6 13 19 21 27 29 TO 32 TABLE ST W36X361
44. 1 2 7 8 10 11 37 49 51 52 TABLE ST W36X393
45. SUPPORTS
46. 24 26 30 TO 34 36 PINNED
47. 38 39 ENFORCED BUT FX FZ MX MY MZ
48. 1 3 FIXED
49. MEMBER RELEASE
50. 6 START MY MZ
51. 19 END MY MZ
52. LOAD 1 LOADTYPE DEAD  TITLE DEAD
53. SELFWEIGHT Y -1
54. LOAD 2 LOADTYPE LIVE  TITLE LIVE_EQUAL
55. JOINT LOAD
56. 5 TO 8 12 15 17 20 FX 1032
57. MEMBER LOAD
58. 8 11 21 27 CON GX 1032 1.25
59. 29 TO 32 CON GX 1032 1.275
60. LOAD 3 LOADTYPE LIVE  TITLE LIVE_BOT
61. MEMBER LOAD
62. 8 11 21 CON GX 1834 1.25
63. 29 30 CON GX 1834 1.28
64. JOINT LOAD
65. 5 6 12 17 FX 1834
66. LOAD COMB 4 DL+LL_EQUAL
67. 1 1.0 2 1.0
68. LOAD COMB 5 DL+LL_BOT
69. 1 1.0 3 1.0
70. LOAD COMB 6 1.4DL+1.4LL_EQUAL
71. 1 1.4 2 1.4
72. LOAD COMB 7 1.4DL+1.4LL_BOT
73. 1 1.4 3 1.4
74. PERFORM ANALYSIS
```

P R O B L E M   S T A T I S T I C S

```
NUMBER OF JOINTS         24  NUMBER OF MEMBERS      28
NUMBER OF PLATES          0  NUMBER OF SOLIDS        0
NUMBER OF SURFACES        0  NUMBER OF SUPPORTS     12
```

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

```
ORIGINAL/FINAL BAND-WIDTH=    15/     7/     39 DOF
TOTAL PRIMARY LOAD CASES =     3, TOTAL DEGREES OF FREEDOM =     108
SIZE OF STIFFNESS MATRIX =        5 DOUBLE  KILO-WORDS
REQRD/AVAIL. DISK SPACE  =     12.1/      0.0 MB
```

```
   75. DEFINE ENVELOPE
   76. 4 5 ENVELOPE 1 TYPE SERVICEABILITY
   77. 6 7 ENVELOPE 2 TYPE STRENGTH
   78. END DEFINE ENVELOPE
   79. PARAMETER 1
   80. CODE BS5950
   81. SGR 1 ALL
   82. LZ 2.21 MEMB 1 2 7 10
   83. LZ 6.1 MEMB 6 19 27
   84. MLT 1 ALL
   85. MX 1 ALL
   86. MY 1 ALL
   87. MYX 1 ALL
   88. CHECK CODE ALL

                      STAAD.Pro CODE CHECKING - (BSI )
                      ***********************

 PROGRAM CODE REVISION V2.13_5950-1_2000
```